CBSE Class 10 Board Exam 2023 – Previous Year Question Paper (Science)

CENTRAL BOARD OF SECONDARY EDUCATION
ALL INDIA SECONDARY SCHOOL EXAMINATION 2023
SUBJECT: SCIENCE (THEORY)

SECTION A (1 Mark Questions)
Q1. A cylindrical conductor of length l and uniform area of cross-section A has resistance R. Another conductor of length 2.5 l and resistance 0.5 R of the same material has area of cross-section:
(a) 5 A (b) 2.5 A (c) 0.5 A (d) 0.2 A

Q2. Assertion (A): The magnetic field lines around a straight current carrying conductor are in the form of concentric circles.
Reason (R): The direction of magnetic field can be found using the Right Hand Thumb Rule.
(a) Both (A) and (R) are true and (R) is correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT correct explanation of (A).
(c) (A) is true, (R) is false. (d) (A) is false, (R) is true.

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SECTION C & D (3 & 5 Mark Questions)

Q15. State Ohm's law. Draw a labelled circuit diagram to verify Ohm's law in the laboratory. Mention two precautions to be taken during the experiment. [3 Marks]

Q16. (a) Derive an expression for the equivalent resistance of three resistors connected in parallel.
(b) An electric lamp of 100 ohms, a toaster of resistance 50 ohms, and a water filter of resistance 500 ohms are connected in parallel to a 220 V source. What is the resistance of an electric iron connected to the same source that takes as much current as all three appliances, and what is the current through it? [5 Marks]